

Mastering IT Complexity

Wireshark in
Industrial Control Systems

Wireshark Meetup

Jasper Rappard

March 2026

KEEP
CALM
and
ANALYZE
PACKETS

LEGIAN



1

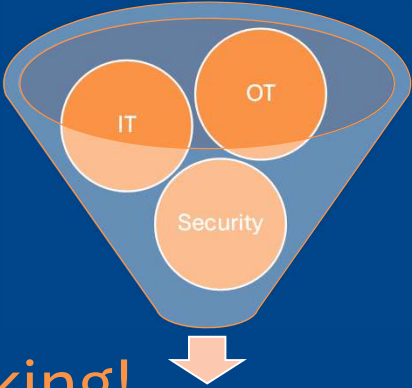
LEGIAN

W – Who am I

Jasper Rappard

<https://www.linkedin.com/in/jasperrp/>

- Legian since 2020
- Wireshark since 2010
- Technical Expert / Security Engineer
- Key skill: Curiosity
- IT & ICS/ OT



Never stop asking!

© Legian 2026

2

LEGIAN

W – What is this session about

In no particular order

- What – What is/ makes ICS/ OT
- Why – Why use wireshark in ICS/ OT
- When – When use Wireshark in ICS/ OT
- Where – Where to capture packets
- Who – Who uses wireshark in ICS/ OT



Q & A – not the best moment to start asking questions...

KEEP CALM and ANALYZE PACKETS

© Legian 2026

3

LEGIAN

W – What is ICS/ OT

Anything cyber/ physical

- You name it...
- Oil/ Gas
- Energy
- Medicine
- Transport & Logistics
- Foot & Beverage
- Smart City – but simple City as well



© Legian 2026

4

W – What makes ICS/ OT

LEGIAN

System of systems

Quality & Efficiency

Deterministic

- Reliablilty
- Predictability
- Continuity
- Safety

Control

Visibility

Tags



© Legian 2026

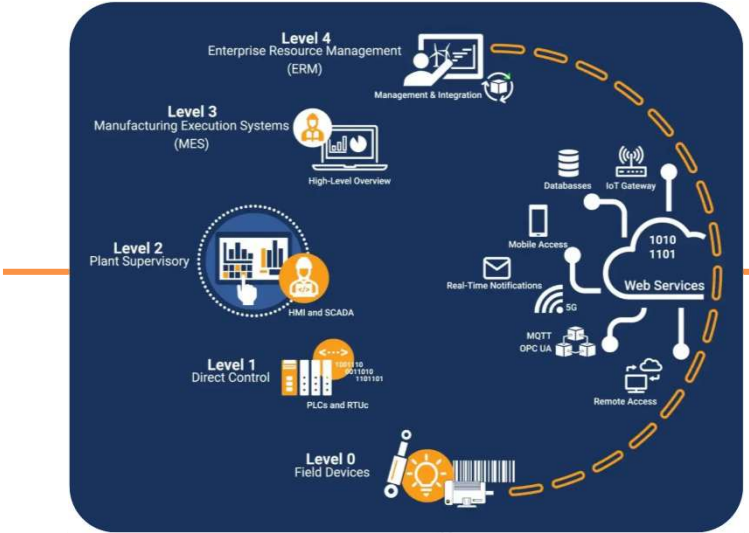
5

W – What makes ICS/ OT

LEGIAN

System of systems

- Connected
- ICS/ OT with IT convergence
- Network traffic
- Protocols



© Legian 2026

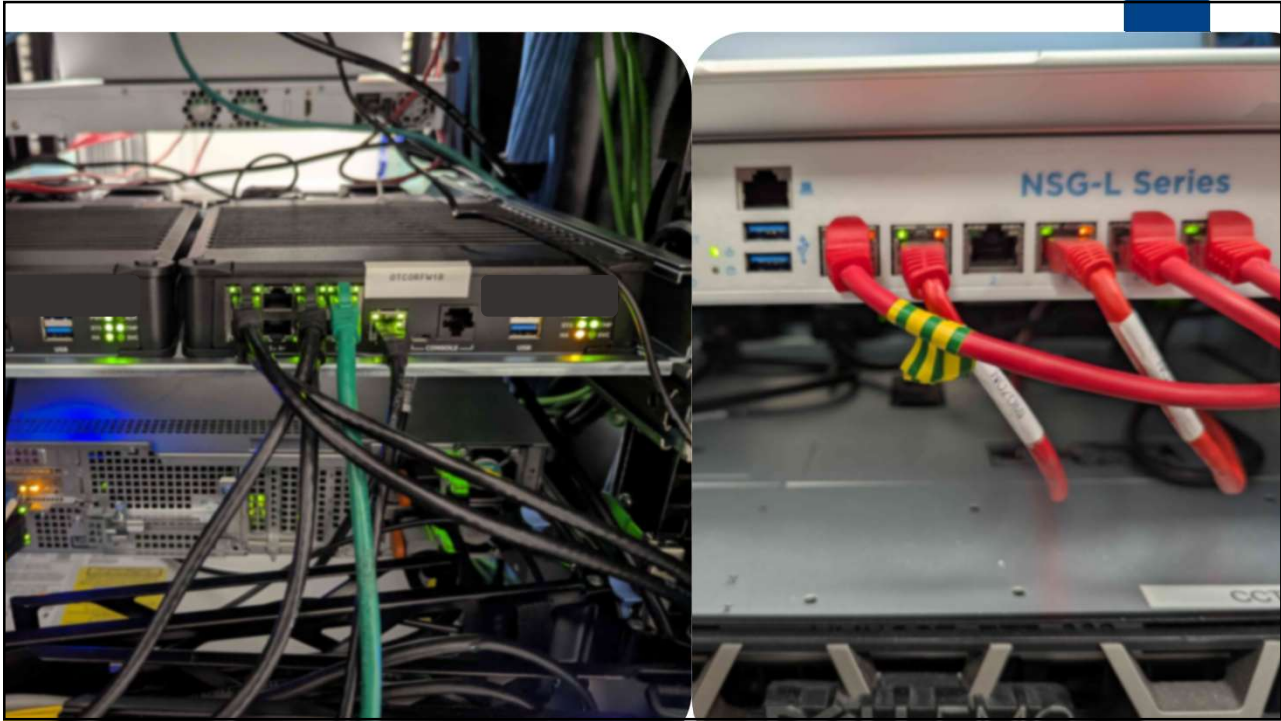
6



7



8



9



10

W – Why use Wireshark in ICS/ OT

LEGIAN

Networks all over the place

- The easy part
- From a structural breakdown it seems easy enough...

CS

ICS network device

ICS device

Supervisory control

Control and monitoring

ICS perimeter security device

ICS switch

Engineering HMI

Operator HMI

Data server

ICS switch

controller

PLC

IO Module

Wireless gateway

Flow sensor

Pressure sensor

Level sensor

Valve

Pump

Sensor

11

W – Why use Wireshark in ICS/ OT

LEGIAN

Choose your packets

- Untill...

Physical or Virtualized Servers

Identity Services

Wireless LAN Controller (WLC)

Active

Standby

Remote Access Server

Core Switches

Cell/Area Zone Levels 0-2

Industrial Zone Levels 0-3 (Plant-wide Network)

Access Switches

Distribution Switch Stack

Level 3 – Site Operations (Control Room)

Camera

Phone

WGB

SSID 5 GHz

WGB

Ethernet/IP

Controller

Controller

Cell/Area Zone – Levels 0-2

LWAP

SSID 2.4 GHz

Stratix 5000/8000 Layer 2 Access Switch

HMI

Soft Starter

Instrumentation

Controller

Drive

I/O

Ethernet/IP

Cell/Area Zone – Levels 0-2

Safety Control I/O

Servo Drive

HMI

Robot

AP

SSID 5 GHz

WGB

Safety I/O

Ethernet/IP

Cell/Area Zone – Levels 0-2

12

LEGIAN - Mastering IT/ OT Complexity

6

Keep in mind – Purdue model

Structured separation of functions

- Secure remote access
- Access to
 - Antivirus updates
 - OS Updates
 - Signaling
 - Management

The diagram illustrates the Purdue model of industrial control systems, structured into five levels:

- Level 5:** Enterprise Network (Enterprise Security Zone)
- Level 4:** E-Mail, Intranet, etc. (Enterprise Security Zone)
- Level 3:** Remote Gateway Services, Patch Management, AV Server, Application Mirror, Web Services Operations, Reverse Proxy, Site Operations (Industrial Demilitarized Zone)
- Level 2:** FactoryTalk Application Server, FactoryTalk Directory, Engineering Workstation, Remote Access Server, Area Supervisory Control (Industrial Security Zone(s))
- Level 1:** FactoryTalk Client, Operator Interface, Engineering Workstation, Operator Interface, Basic Control (Cell/Area Zone(s))
- Level 0:** Batch Control, Discrete Control, Drive Control, Continuous Process Control, Safety Control (Cell/Area Zone(s))

Additional components shown include Sensors, Drives, Actuators, Robots, and Process.

© Legian 2026

13

W – Why use Wireshark in ICS/ OT

Choose your packets

- ICS networks** require visibility
 - What is in there
 - Who does what
 - Is it all running as it should
- Wireshark** supports
 - Baselining
 - Architecting/ re-engineering
 - Troubleshooting
 - Security

The image shows a network analysis setup. A network switch is connected to a computer running Wireshark. The Wireshark interface is visible on the monitor, displaying network traffic. The Wireshark logo is prominently displayed in the center of the image.

© Legian 2026

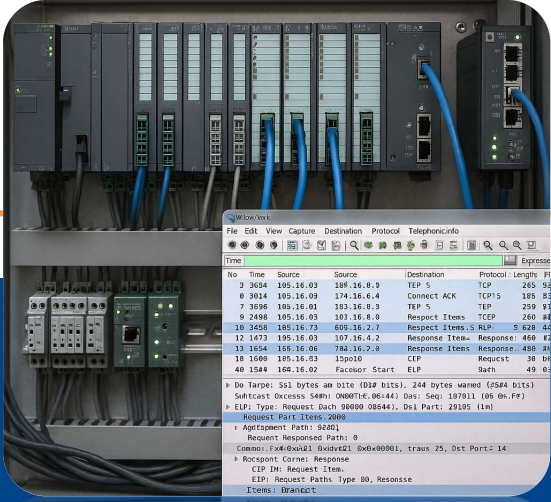
14

W – When use Wireshark in ICS/ OT

W – Who uses Wireshark in ICS/ OT

Usecases & Stakeholders

- **Asset discovery**
 - Asset Owner (Plant manager)
 - Infra engineering Team
- **Baseline**
 - Maintenance Team
 - Security Operations
- **Re-engineering**
 - Infra engineering team
 - Security Architect
- **Incident response**
 - Security Operations



© Legian 2026

15

W – Wireshark in ICS/ OT

Use it safely

- Passive only
- No installation on critical devices
- Offline analysis preferred
- Coordinate with operations
- Avoid active scanning



© Legian 2026

16

W – Where to capture

SPAN vs TAP

SPAN

- Non-intrusive
- Flexible

Use in *production*

- Easy to configure
- Live traffic
- Filtering (often vlan)

TAP

- Intrusive
- Fixed

Use in *FAT*

- Factory Acceptance Test
- Real life scenario

© Legian 2026

17

W – Where to capture

Windows devices – no Wireshark needed

No need to install dumpcap/ wireshark

- **NETSH trace**
 - Includes CAB file with system info (including Process IDs)
 - External ETL2PCAPNG
 - PCAPNG includes Process IDs)
- **PKTMON**
 - Bare network capture
 - Native ETL2PCAP

Destination	Protocol	Comment	Info
68.5 172.16.182.85	ICMP, HSPerContract	TID=1336 Echo (ping) request	1d4b083c, seq=36329/
68.5 172.16.99.12	ICMP, HSPerContract	TID=4184 Echo (ping) request	1d4b0808, seq=36329/
99.12 172.16.68.5	ICMP, HSPerContract	TID= Echo (ping) reply	1d4b0808, seq=36329/
182.85 172.16.68.5	ICMP, HSPerContract	TID= Echo (ping) reply	1d4b083c, seq=36329/
68.5 172.16.99.12	ICMP, HSPerContract	TID=1336 Echo (ping) request	1d4b0808, seq=36330/
68.5 172.16.182.85	ICMP, HSPerContract	TID=1336 Echo (ping) request	1d4b083c, seq=36330/
99.12 172.16.68.5	ICMP, HSPerContract	TID= Echo (ping) reply	1d4b0808, seq=36330/
182.85 172.16.68.5	ICMP, HSPerContract	TID= Echo (ping) reply	1d4b083c, seq=36330/
68.5 172.23.41.4	TLSv1.2	TID=2988 Application Data	
41.4 172.16.68.5	TCP	TID= TID= 23568 + 57934 [ACK] Seq=1 Ack=305 Win=1638	
68.5 172.23.41.4	TLSv1.2	TID=2988 Application Data	
41.4 172.16.68.5	TCP	TID= TID= 23568 + 57934 [ACK] Seq=1 Ack=301 Win=1634	
68.5 172.16.67.18	TLSv1.2	TID=1888 Application Data	
67.18 172.16.68.5	TCP	TID= TID= 56928 + 3389 [ACK] Seq=1 Ack=52 Win=8288 L	
68.5 182.168.8.58	ICMP, HSPerContract	TID=279 TID=2912 Echo (ping) request	1d4b275a, seq=36329/
1.8.58 172.16.68.5	ICMP, HSPerContract	TID= TID= Echo (ping) reply	1d4b275a, seq=36329/
68.5 182.168.8.58	ICMP, HSPerContract	TID=279 TID=2912 Echo (ping) request	1d4b275a, seq=36329/
1.8.58 172.16.68.5	ICMP, HSPerContract	TID= TID= Echo (ping) reply	1d4b275a, seq=36329/
68.5 182.168.8.58	ICMP, HSPerContract	TID=279 TID=2912 Echo (ping) request	1d4b275a, seq=36329/
1.8.58 172.16.68.5	ICMP, HSPerContract	TID= TID= Echo (ping) reply	1d4b275a, seq=36329/
68.5 182.168.8.58	ICMP, HSPerContract	TID=279 TID=2912 Echo (ping) request	1d4b275a, seq=36330/
1.8.58 172.16.68.5	ICMP, HSPerContract	TID= TID= Echo (ping) reply	1d4b275a, seq=36330/
68.5 172.23.41.4	TLSv1.2	TID=2988 Application Data	
41.4 172.16.68.5	TCP	TID= TID= 23568 + 57934 [ACK] Seq=1 Ack=757 Win=1638	
68.5 172.16.99.13	TCP	TID= TID= 56186 + 445 [ACK] Seq=1 Ack=1 Win=1823 Len	
99.13 172.16.68.5	TCP	TID= TID= 445 + 56186 [ACK] Seq=1 Ack=1 Win=6216 Len	
68.5 172.16.182.39	SNMP	TID=1252 get-request 1.3.6.1.2.1.25.1.1.8	
68.5 182.168.8.58	SNMP	TID=1248 get-request 1.3.6.1.2.1.31.1.1.3.6.7 1.3.6	
68.5 182.168.8.57	SNMP	TID=1256 get-request 1.3.6.1.2.1.31.1.1.3.6.7 1.3.6	
181.39 172.16.68.5	SNMP	TID= TID= get-response 1.3.6.1.2.1.25.1.1.8	
68.5 172.23.41.4	TLSv1.2	TID=2988 Application Data	
6.55 172.16.68.5	SNMP	TID= TID= get-response 1.3.6.1.2.1.31.1.1.3.6.7 1.3	
6.57 172.16.68.5	SNMP	TID= TID= get-response 1.3.6.1.2.1.31.1.1.3.6.7 1.3	
41.4 172.16.68.5	TCP	TID= TID= 23568 + 57934 [ACK] Seq=1 Ack=841 Win=1638	
68.5 172.23.41.4	TLSv1.2	TID=2988 Application Data	
41.4 172.16.68.5	TCP	TID= TID= 23568 + 57934 [ACK] Seq=1 Ack=1277 Win=1638	
6.5 172.23.41.4	TLSv1.2	TID=2988 Application Data	

© Legian 2026

18

LEGIAN - Mastering IT/ OT Complexity

9

W – What to use to get packets

LEGIAN

NETSH trace – Process ID in packet comment

The screenshot shows the Wireshark interface with a packet capture list on the left and packet details on the right. A red box highlights a packet comment in the list, which contains a process ID. The details pane on the right shows the packet's structure, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol.

© Legian 2026

19

W – When use Wireshark in ICS/ OT

LEGIAN

Usecases

- **Asset discovery**
 - Sit back and relax
 - Discovery protocols all over the place
- How to get to the packets?
 - SPAN
 - IDS
 - Firewall
 - Servers

The diagram illustrates the process of using Wireshark for asset discovery in an ICS/OT environment. It shows a network topology with a SCADA Server, a PLC, and a SPAN/Mirror Port. Wireshark is connected to the network, capturing traffic. The captured data is processed to extract metadata, including Device Type, OS, IP/MAC, and Vendor. This information is then stored in an Asset Inventory, which lists assets like PLcc, OS, IP/MAC, Vendor C, and Vendor B.

© Legian 2026

20

Cisco Discovery Protocol

No.

Time

Source

Destination

Protocol

Info

1

0.000000

Cisco_18:9a:41

CDP/VTP/DTP/P...

CDP

Device ID: myswitch Port ID: FastEthernet0/1

IEEE 802.3 Ethernet

Logical-Link Control

Cisco Discovery Protocol

Version: 2

TTL: 180 seconds

Checksum: 0x09a0 [correct]

[Checksum Status: Good]

Device ID: myswitch

Type: Device ID (0x0001)

Length: 12

Device ID: myswitch

Addresses

Port ID: FastEthernet0/1

Capabilities

Type: Capabilities (0x0004)

Length: 8

Capabilities: 0x00000028

.....0 = Router: No

.....0 = Transparent Bridge: No

.....0 = Source Route Bridge: No

.....1 = Switch: Yes

.....0 = Host: No

.....1 = IGMP capable: Yes

.....0 = Repeater: No

.....0 = VoIP Phone: No

.....0 = Remotely Managed Device: No

.....0 = CVTA/STP Dispute Resolution/Cisco VT Camera: No

.....0 = Two Port Mac Relay: No

Software Version

Type: Software version (0x0005)

Length: 276

Software version: Cisco Internetwork Operating System Software

Software version: IOS (tm) C2950 Software (C2950-IGK2L2Q4-M), Version 12.1(22)EA14, RELEASE SOFTWARE (fc1)

Software version: Technical Support: <http://www.cisco.com/techsupport>

Software version: Copyright (c) 1986-2010 by cisco Systems, Inc.

Software version: Compiled Tue 26-Oct-10 10:35 by nburra

Platform: cisco WS-C2950-12

Native VLAN: 1

Duplex: Full

Trust Bitmap: 0x00

Untrusted port CoS: 0x00

Management Addresses

Type: Management Address (0x0016)

Length: 17

Number of addresses: 1

IP address: 192.168.0.253

© Legian 2026

21

W – When use Wireshark in ICS/ OT

Usecases

Re-engineering

Find flat networks during asset discovery

Large broadcast domains

Architecture review

Segmentation as security measure

Better security means better business

Continuity/ reliability

Health, Safety & Environment

POWER SUPPLY

CPU

24V DC INPUT

0 1 2 3 4 5 6 7

8 9 10 11 12 13 14 15

16 17 18 19 20 21 22 23

24 25 26 27 28 29 30 31

Ethernet

L1

L2/N

Temp

Speed

+24V 0V

Tag values to HMI

Tag values from HMI

© Legian 2026

22

W – When use Wireshark in ICS/ OT

LEGIAN

Usecases

- Baseline
 - Know the normal
 - Continuous monitoring for anomalies
- Start capturing BEFORE start-up
 - Configuration (Download to PLC)
 - Initialization
 - Run
 - Shutdown scenarios
 - Maintenance
 - Safety

© Legian 2026

23

W – When use Wireshark in ICS/ OT

LEGIAN

Usecases

- Incident response
 - Troubleshooting
 - Cyber incident
 - Complementary to IDS

© Legian 2026

24

Examples of traffic

A typical day on the plant...

- Profinet
- LLDP
- CDP
- Spanning Tree
- IPv6
- S7 – over COTP
- TDS – database traffic
- Modbus TCP
- CIP – over EtherNet/IP
- DCE/RCP
- IGMP

Protocol

- Frame
 - Ethernet
 - PROFINET Real-Time Protocol
 - PROFINET DCP
 - Logical-Link Control
 - Spanning Tree Protocol
 - Dynamic Trunk Protocol
 - Cisco Discovery Protocol
 - Link Layer Discovery Protocol
 - Internet Protocol Version 6
 - User Datagram Protocol
 - Link-local Multicast Name Resolution
 - DHCPv6
 - Internet Control Message Protocol v6
 - Internet Protocol Version 4
 - User Datagram Protocol
 - Simple Service Discovery Protocol
 - NetBIOS Name Service
 - Link-local Multicast Name Resolution
 - Distributed Computing Environment / Remote Procedure Call (DCE/RPC)
 - RCG (pidl)
 - Data
 - Transmission Control Protocol
 - Transport Layer Security
 - TPKT - ISO on TCP - RFC1006
 - ISO 8073/X.224 COTP Connection-Oriented Transport Protocol
 - S7 Communication
 - Tabular Data Stream
 - Malformed Packet
 - Socks Protocol
 - SICK CoLA A
 - NetBIOS Session Service
 - SMB2 (Server Message Block Protocol version 2)
 - SMB (Server Message Block Protocol)
 - Modbus/TCP
 - Modbus
 - EtherNet/IP (Industrial Protocol)
 - Common Industrial Protocol
 - CIP Connection Manager
 - Distributed Computing Environment / Remote Procedure Call (DCE/RPC)
 - DCOM OXID Resolver
 - Data
 - Internet Group Management Protocol
 - Internet Control Message Protocol
 - Address Resolution Protocol
 - Cisco ISL

© Legian 2026

25

Siemens utilizing LLDP & Profinet

Cisco TIA portal – uses LLDP and Profinet to manage plant systems

© Legian 2026

26

LLDP

> Ethernet II, Src: SiemensIndus_34:21:d5 (e0:dc:a0:34:21:d5), Dst: Nearest-Bridge (01:80:c2:00:00:0e)

> Link Layer Discovery Protocol

> Chassis Subtype = Locally assigned, Id: S7-1200

> Port Subtype = Locally assigned, Id: port-001.plcxbd0ed

> Time To Live = 20 sec

> Port Description = Siemens, SIMATIC S7, Ethernet Port, X1 P1

> System Description = Siemens, SIMATIC S7, CPU-1200, 6ES7 214-1AG40-0XB0 HW: 7, FW: V.4.

> Capabilities

> Management Address

> PROFIBUS Nutzerorganisation e.V. - Port Status

> PROFIBUS Nutzerorganisation e.V. - Chassis MAC

> Ieee 802.3 - MAC/PHY Configuration/Status

> Ieee 802.3 - Maximum Frame Size

> End of LLDPDU

Management Address

0001 000. = TLV Type: Management Address (8)

.... ..0 0001 0100 = TLV Length: 20

Address String Length: 5

Address Subtype: IPv4 (1)

Management Address: 192.168.0.1

Interface Subtype: ifIndex (2)

Interface Number: 1

OID String Length: 8

Object Identifier: 1.3.6.1.4.1.24686 (iso.3.6.1.4.1.24686)

© Legian 2026



27

LLDP

> Ethernet II, Src: SiemensIndus_29:c6:e7 (e0:dc:a0:29:c6:e7), Dst: Nearest-Bridge (01:80:c2:00:00:0e)

> Link Layer Discovery Protocol

> Chassis Subtype = Locally assigned, Id: hmixb110d0

> Port Subtype = Locally assigned, Id: port-001

> Time To Live = 20 sec

> Port Description = Siemens, SIMATIC S7, Ethernet Port, X1 P1

> System Description = Siemens, SIMATIC HMI, KTP700 Basic 6AV2 123-2GB0

> Capabilities

> Management Address

> PROFIBUS Nutzerorganisation e.V. - Port Status

> PROFIBUS Nutzerorganisation e.V. - Chassis MAC

> Ieee 802.3 - MAC/PHY Configuration/Status

> Ieee 802.3 - Maximum Frame Size

> End of LLDPDU

Management Address

0001 000. = TLV Type: Management Address (8)

.... ..0 0001 0100 = TLV Length: 20

Address String Length: 5

Address Subtype: IPv4 (1)

Management Address: 192.168.0.2

Interface Subtype: ifIndex (2)

Interface Number: 2

OID String Length: 8

Object Identifier: 1.3.6.1.4.1.24686 (iso.3.6.1.4.1.24686)

© Legian 2026



28

LEGIAN - Mastering IT/ OT Complexity

14

TIA Portal set IP address via Profinet DCP

Profinet Discovery & Configuration

Source	Destination	Protocol	IP Address	Info
VMware_ba:09:ea	Siemens_93:cf:32	PN-DCP		Ident Req, Xid:0x1000001, All
Siemens_93:cf:32	VMware_ba:09:ea	PN-DCP	192.168.0.6	Ident Ok, Xid:0x1000001, Dev-Options(13), DeviceVendorValue, NameOfStation:"X208-BORD", Dev-ID, Dev-Role, IP
VMware_ba:09:ea	Siemens_93:cf:32	PN-DCP	192.168.0.10	Set Req, Xid:0x1000001, IP
Siemens_93:cf:32	VMware_ba:09:ea	PN-DCP		Set Ok, Xid:0x1000001, Response(Ok)
Siemens_93:cf:32	Broadcast	ARP		Who has 192.168.0.10? Tell 192.168.0.6
Siemens_93:cf:32	Broadcast	ARP		Gratuitous ARP for 192.168.0.10 (Reply)

Switch_1 [SCALANCE XB208]

General IO tags System constants

PROFINET interface [X1]

Management

Module parameters

System

Layer 2

Layer 3

Security



© Legian 2026

29

MQTT

Plain text - Often seen in IoT environments

No.	Time	Source	Destination	Protocol	Info
1	0.000000	192.168.1.131	35.156.122.117	TCP	28465 → 1883 [SYN] Seq=0 Win=0 Len=0
2	0.074311	35.156.122.117	192.168.1.131	TCP	1883 → 28465 [SYN, ACK] Seq=0 Win=0 Len=0
3	0.074416	192.168.1.131	35.156.122.117	TCP	28465 → 1883 [ACK] Seq=1 Win=0 Len=0
4	0.077265	192.168.1.131	35.156.122.117	MQTT	Connect Command
5	0.150169	35.156.122.117	192.168.1.131	TCP	1883 → 28465 [ACK] Seq=1 Win=0 Len=0
6	0.165372	35.156.122.117	192.168.1.131	MQTT	Connect Ack
7	0.215125	192.168.1.131	35.156.122.117	TCP	28465 → 1883 [ACK] Seq=38 Win=0 Len=0
8	3.866576	192.168.1.131	35.156.122.117	MQTT	Subscribe Request (id=1) [/sub]
9	3.944973	35.156.122.117	192.168.1.131	MQTT	Subscribe Ack (id=1)
10	3.999711	192.168.1.131	35.156.122.117	TCP	28465 → 1883 [ACK] Seq=49 Win=0 Len=0
11	10.613791	192.168.1.131	35.156.122.117	MQTT	Publish Message [/sub]
12	10.689175	35.156.122.117	192.168.1.131	MQTT	Publish Message [/sub]
13	10.735208	192.168.1.131	35.156.122.117	TCP	28465 → 1883 [ACK] Seq=66 Win=0 Len=0
14	13.855541	192.168.1.131	35.156.122.117	MQTT	Publish Message [/sub]
15	13.937938	35.156.122.117	192.168.1.131	MQTT	Publish Message [/sub]

Frame 4: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits)

Ethernet II, Src: Intel_f4:7d:29 (00:19:d2:f4:7d:29), Dst: TpLinkTechno_96:df:01 (c4:e9:84:96:df:01)

Internet Protocol Version 4, Src: 192.168.1.131, Dst: 35.156.122.117

Transmission Control Protocol, Src Port: 28465, Dst Port: 1883, Seq: 1, Ack: 1, Len: 37

MQ Telemetry Transport Protocol, Connect Command

Header Flags: 0x10, Message Type: Connect Command

Msg Len: 35

Protocol Name Length: 4

Protocol Name: MQTT

Version: MQTT v3.1.1 (4)

Connect Flags: 0x02, User Name Flag, Password Flag, QoS Level: At most once delivery (Fire and Forget), Clean Session: 0

Keep Alive: 60

Client ID Length: 11

Client ID: MQTT_Client

User Name Length: 4

User Name: demo

Password Length: 4

Password: demo

Wireshark - Follow TCP Stream (tcp.stream eq 0) - mqtt_user_credentials.pcap

...MQTT...MQTT_Client...demo...demo

.../sub...

0.../subtest data

0.../subtest data

0.../subtest data x

0.../subtest data y

0.../subtest data y

0.../subtest data z

0.../subtest data z

...

30

LEGIAN - Mastering IT/ OT Complexity

15

31

32

Take aways

LEGIAN

ATTACK the plant – loss of view & control

- Yes, but...
- Putting PLC in RUN mode (switch or key-switch)
 - Prevents ability to download program
 - Prevents ability to remotely control the PLC

BUT

Does NOT prevent data to be altered (spoofing attack) on the network



© Legian 2026

33

Q & A

LEGIAN

Not the best moment to start asking questions...

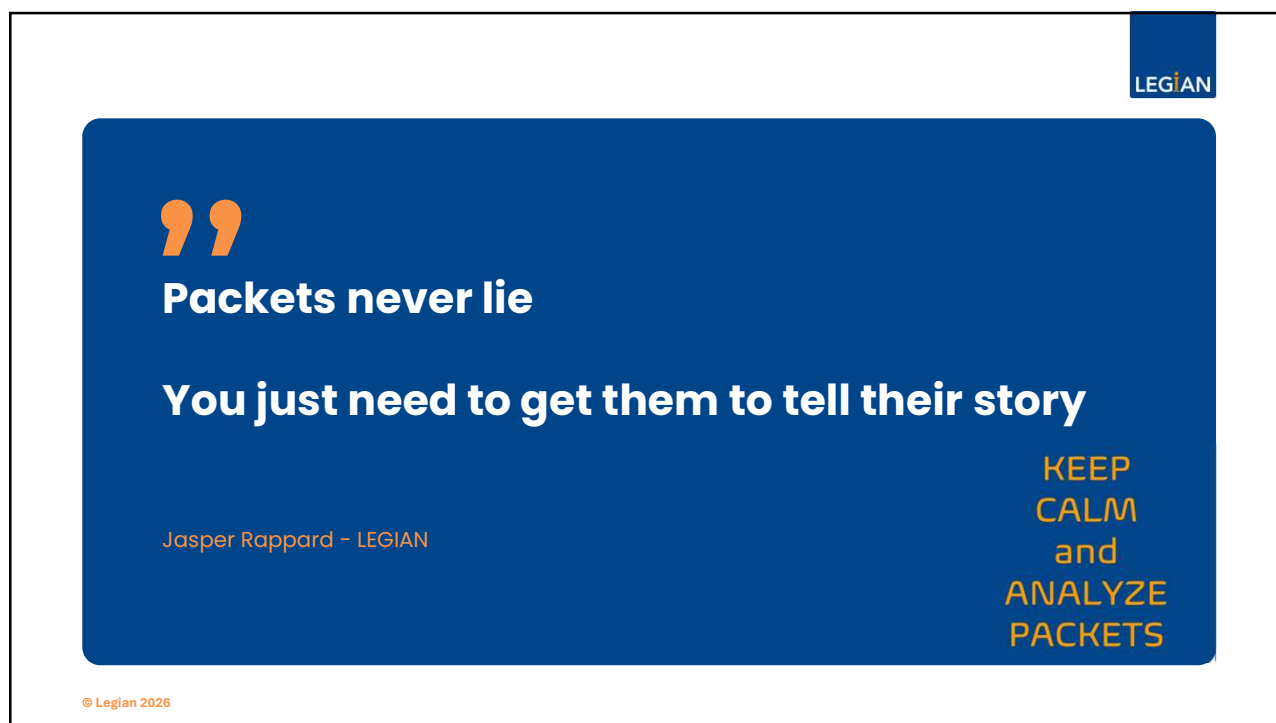
ICS is a lot...

ICS is fun

Even more fun with Wireshark

© Legian 2026

34



LEGIAN

”

Packets never lie

You just need to get them to tell their story

Jasper Rappard - LEGIAN

KEEP
CALM
and
ANALYZE
PACKETS

© Legian 2026

35

On a personal note

Wireshark foundation – development, maintenance & support

Over the last 10 years I have gratefully used Wireshark to help companies overcome issues with their systems. And have fun in learning a lot about networks, systems and software.

Going *deep*, questioning *assumptions* and providing *irrefutable evidence* would not have been possible without the ongoing endeavors of the Wireshark team – Wireshark Foundation.

As rightfully stated by Kasper Jensen, Open Source is no freeware. It is hard work by passionate people. Often in their spare time, often without guarantees for sustained efforts.

I love Wireshark, as it helps me to do what I love. Mastering complexity, by breaking down events into what really happens on the network.

Packets never lie, and Wireshark helps me making them telling their story...

I support the Wireshark Foundation – hopefully, you do to. Every penny counts.

Consider donating: <https://wiresharkfoundation.org/donate>

[Kasper Jensen post on LinkedIn - https://tinyurl.com/KasperJensenOpenSource](https://tinyurl.com/KasperJensenOpenSource)

36